

Select and construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information

Learning intention

- select and construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information.

Teach and model

- using spreadsheets to aid the presentation and analysis of data.
- distinguishing between discrete and continuous data and selecting appropriate data representations.
- constructing food webs to represent feeding relationships and flows of energy and matter in an ecosystem.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.